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TECHNOLOGICAL UNIVERSITY DUBLIN
CITY CAMPUS - GRANGEGORMAN

Master of Science in Computer Science
(Advanced Software Development - Full-Time)

Master of Science in Computer Science
(Data Science - Part-Time)

Year 1

SUPPLEMENTAL
EXAMINATIONS 2023/24

DATA MINING

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Two Hours

INSTRUCTIONS TO CANDIDATES

Answer all parts in Question 1 and 2. And in question 3 answer 4 out of 5 parts.

1. Answer the following questions: (30 marks)

(a) Explain what is Natural Language Processing and why is it so important, including 3 examples of some applications

- **Natural Language Processing (NLP)** is a field of artificial intelligence that focuses on the interaction between computers and human language. (6 marks)
- **Main difficulties are (Up to 4 marks):**
 1. **Ambiguity:** Language is often ambiguous, with words or phrases having multiple meanings. (2 marks)
 2. **Context:** Understanding context and nuances in language is challenging. (2 marks)
 3. **Data Variability:** Language data varies widely in structure, dialects, and styles. (2 marks)
 4. **Sarcasm/Irony:** Detecting sarcasm and irony can be complex due to their non-literal nature. (2 marks)

(10 marks)

(b) Discuss what is Deep Learning and why has it become so relevant in the last few years.

- **Deep Learning** is a subset of machine learning that uses multi-layered neural networks to process complex data. It's become highly relevant due to advances in hardware, datasets, and network architectures, enabling its application in various fields. (6 marks)
- The recent relevance of deep learning is attributed to enhanced hardware, data, and network designs. These developments have led to its widespread use in areas like healthcare, autonomous vehicles, and natural language processing. (4 marks)

(10 marks)

(c) Discuss the concept of Entropy and its relationship with Information Gain.

- Entropy is a measure of disorder or impurity in a dataset. (4 marks)
- Information Gain, on the other hand, is a metric used in decision tree algorithms to evaluate the effectiveness of splitting a dataset based on a particular attribute. It measures the reduction in entropy achieved by dividing the dataset with respect to that attribute. In simpler terms, Information Gain quantifies how much knowing the outcome of an attribute reduces our uncertainty (entropy) about the final classification. (6 marks)

(10 marks)

2. Answer the following three points: (30 marks)

- (a) Explain the concept of clustering including the two typical examples K-Means and hierarchical clustering.

- Clustering is a data mining technique that involves grouping similar data points together based on their inherent characteristics or features. It's used to discover patterns or structures within data. (4 marks)
- Two typical examples of clustering methods are:
 - K-Means Clustering: K-Means aims to partition data points into 'K' distinct clusters where each data point belongs to the cluster with the nearest mean. It uses centroids as the center points of clusters and iteratively adjusts them to minimize the distance between data points and their assigned centroids. (3 marks)
 - Hierarchical Clustering: Hierarchical clustering builds a tree-like structure (dendrogram) where data points start as individual clusters and are progressively merged into larger ones. It provides a visual representation of how data points are related to each other, and you can decide on the number of clusters by cutting the dendrogram at a certain level. (3 marks)

(10 marks)

- (b) Discuss the difference between filters and wrapper methods including when is best using one or the other.

Filter methods select features based on statistical measures like correlation (3 marks), while wrapper methods use model performance as a criterion for feature selection (3 marks). Filter methods are computationally less expensive and are recommended for high-dimensional data (2 marks). Wrapper methods are computationally expensive but can lead to better model performance (2 marks).

(10 marks)

- (c) Give the typical steps you will need to create a pipeline for sentiment analysis using data mining techniques.

To create a sentiment analysis pipeline using data mining techniques, follow these steps: Collect text data with sentiment labels (1.5 marks), preprocess the text by cleaning and transforming it (1.5 marks), extract numerical features from the text (1.5 marks), choose an appropriate model (1.5 marks), train and evaluate the model, apply it to new data (1.5 marks), interpret the results (1.5 marks), deploy the model if it performs well and continuously monitor and retrain it to maintain accuracy and relevance (1 mark).

(10 marks)

3. Answer four out of the following five questions: (40 marks)

- (a) Discuss the purpose of Principal Component Analysis (PCA) in data analysis, and when do you recommend to use it.

- **Principal Component Analysis (PCA)** is a dimensionality reduction technique used to simplify complex datasets while preserving essential information. It works by finding new orthogonal axes (principal components) in the data that maximize variance. These components help reduce the dimensionality of the data while retaining the most significant variance. (6 marks)
- PCA is recommended for high-dimensional datasets where reducing complexity is essential, particularly in fields like image processing, genomics, and finance. It improves computational efficiency, mitigates overfitting, and uncovers hidden patterns within the data, making it a valuable tool for exploratory data analysis and modeling. (4 marks)

(10 marks)

- (b) Discuss when the following feature selection methods should be applied: Filters, Wrappers, Deep Learning and Principal Component Analysis.

- **Filters:** Apply when dealing with large datasets to quickly eliminate irrelevant or redundant features for improved efficiency.
- **Wrappers:** Use when optimizing feature selection for a specific machine learning model, particularly for maximizing predictive performance.
- **Deep Learning:** Ideal for complex, high-dimensional data like images or sequences, where deep neural networks can automatically learn relevant features.
- **Principal Component Analysis (PCA):** Employ for high-dimensional data to reduce dimensionality and enhance efficiency, especially in exploratory data analysis and visualization.

(10 marks)

- (c) Discuss the data preparation step in the the CRISP-DM process and give examples of two common techniques.

Main steps for CRISP-DM:

- **Data Preparation:** Clean and preprocess the data. Select relevant features and transform the dataset (6 marks).
- **Examples (Up to 4 marks):** Replacing missing values, Data aggregation, normalization, data balance, etc.

(10 marks)

- (d) Discuss the concept of Reinforcement Learning and why does it get good results in games like Atari.

- **Reinforcement Learning (RL)** is a machine learning approach where an agent learns by taking actions in an environment to maximize rewards. It involves the agent's policy, actions, and rewards, aiming to find the best strategy for long-term gain. (5 marks)
- RL is particularly effective in game environments like Atari because of its ability to adapt and learn optimal strategies without explicit human guidance. In these games, the agent interacts with the environment, receives feedback in the form of rewards (score increases or decreases), and continually refines its decision-making process to achieve high scores. (5 marks)

(10 marks)

- (e) Give short definitions for the concepts related to explainability: Accountability, Trustworthiness, Confidence, Causality, and "Fairness and ethical decision".

- **Accountability:** Holding individuals or systems responsible for their actions and decisions, ensuring transparency and traceability in the decision-making process. (2 marks)
- **Trustworthiness:** The quality of being reliable and dependable, indicating users can have confidence in the system's performance and results. (2 marks)
- **Confidence:** The level of certainty or trust in the accuracy and reliability of a system's predictions, often expressed as a probability or confidence score. (2 marks)
- **Causality:** The relationship between cause and effect, exploring how one event or variable influences another, crucial for AI explainability. (2 marks)
- **Fairness and Ethical Decision:** Ensuring that AI decisions align with ethical principles, do not discriminate, and promote equitable outcomes in accordance with societal values. (2 marks)

(10 marks)